

SmartRouteADMM - AI-driven Bus Scheduling and Route Management System using ADMM

Final Review Presentation

TEAM - 07

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Problem Statement

Urban bus systems in densely populated areas face significant challenges in adapting to fluctuating passenger demand across different time periods, such as peak versus offpeak hours or weekdays versus holidays. Traditional static and pre-planned scheduling approaches fail to address real-time variations, leading to issues like overcrowded buses, underutilized capacity, long waiting times, and inefficient routing. These inefficiencies result in reduced service quality, passenger dissatisfaction, and increased operational costs. There is a critical need for a dynamic, automated, and demand-responsive bus management framework that can optimize routing, scheduling, and capacity utilization in real-time to enhance operational efficiency and passenger satisfaction.

- **Dynamic Bus Scheduling and Routing :** Develop a demand-responsive bus management system that adapts to real-time or predicted passenger demand, ensuring efficient routing and scheduling.
- **Capacity Optimization:** Maximize the utilization of available bus capacity to address issues like overcrowding and underutilized buses.
- **Operational Cost Reduction:** Minimize operational costs by optimizing resource allocation and reducing delays
- **Passenger Satisfaction:** Improve service quality by reducing waiting times, enhancing reliability, and ensuring a better travel experience for commuters.

Literature Review

Efficient bus routing and scheduling have been extensively studied to address urban transportation challenges such as traffic congestion, fluctuating passenger demand, and operational cost optimization.

Study/Authors	Method Used	Focus Area	Key Outcomes
Prof. Priyanka Salunkhe et al.	Automated Scheduling System	Real-time scheduling, passenger-centric services	Improved route allocation, reduced delays, better timetable adherence
Cao and Ceder	Genetic Algorithm (GA)	Real-time rescheduling	Minimized total travel time and reduced number of active buses
Liu et al.	Genetic Algorithm (GA)	Traffic disruption scenarios	Reduced waiting time, in-vehicle time, and operational delays
Wang et al.	Genetic Algorithm (GA)	Congestion scenarios	Reduced operational costs under heavy traffic conditions
Li et al.	Auction-Based Algorithm	Dynamic schedule adjustment during disruptions	Minimized service delays while ensuring fair resource allocation
Nikolić and Teodorović	Bee Colony Optimization	Demand fluctuation management	Balanced supply-demand and reduced transportation costs

Research Gaps Identified

- **Lack of Real-Time Adaptability:** Most existing bus scheduling systems are static or semi-dynamic, making them ineffective in responding to real-time fluctuations in passenger demand.
- **Centralized Optimization techniques:** Traditional centralized optimization approaches become computationally inefficient and non-scalable for large urban transit networks.
- **Avoidance of Dynamic constraints:** Most current methods do not adequately consider practical constraints like bus capacity, driver availability, or emergency backup vehicle deployment.

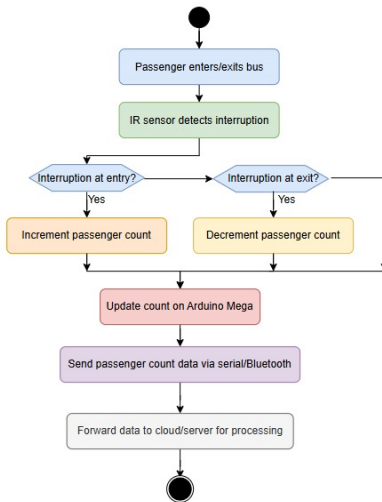
Novelty of the Proposed Work

- **AI-Powered Passenger Demand Forecasting:** Uses LSTM-based deep learning models trained on synthetic yet realistic data to forecast short-term passenger flow accurately.
- **ADMM-Based Optimization:** Applies Alternating Direction Method of Multipliers to split complex optimization into subproblems, enabling real-time, scalable computation.
- **Resource-Aware Scheduling & Backup Handling:** Incorporates real-world constraints including backup bus dispatching and driver allocation, rarely addressed in earlier systems.
- **Scalable Architecture:** Designed for extensibility across urban and semi-urban transit networks with flexible routing, node addition, and cloud-based control.

The proposed system for demonstration purpose of our idea is composed of four main components:

- **IR Sensor Network** for real-time passenger count so that we could identify fluctuation of passenger demand at various bus stops.
- **Model Prediction** using LSTM model for predicting the number of passengers of the next day .so that the scheduling can be done based on passenger count.
- **ADMM Optimization** for the computation of decentralized and to find the most optimized routes.
- **Dynamic Bus Dispatching and Communication System** for real-time implementation where all the logic of the code is running in the backend and an user friendly display to view the schedules.

System Architecture



Implementation

Hardware Setup Using Arduino Mega and IR sensors

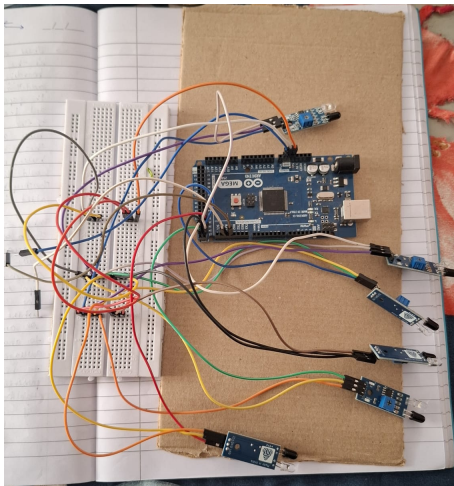


Figure: Hardware Setup

System Flow for SmartRouteADMM

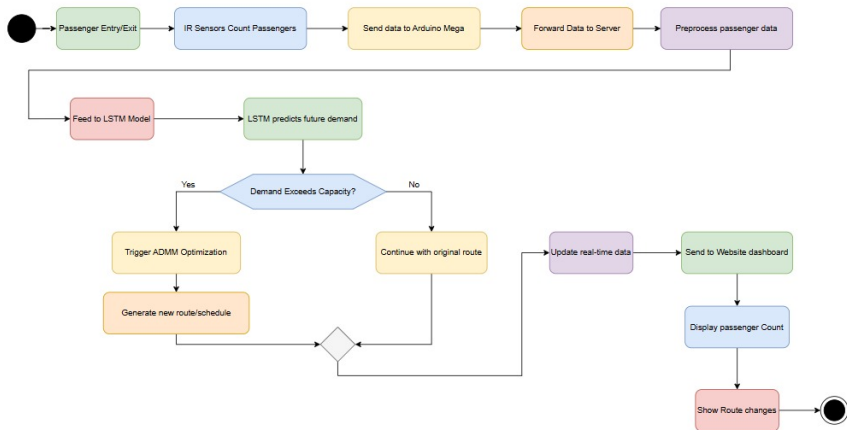


Figure: Overall Implementation Flow of SmartRouteADMM System

LSTM Model Training and Evaluation

- An LSTM model was trained on synthetically generated time-series data to forecast passenger demand at bus stops. Over 50 epochs, the model showed decreasing training and validation loss, converging around 0.0332 and 0.0336, respectively, indicating good generalization

Results and Discussion



Figure: LSTM Training and Validation Loss

Results and Discussion

```
[DEBUG] ADMM Input - Routes: {'Route A': [0, 1], 'Route B': [2, 3], 'Route C': [4, 5]}
[DEBUG] ADMM Input - Total Buses: 10
[DEBUG] ADMM Input - Distances: [2.5, 3.0, 1.8, 2.2, 3.5, 1.0]
[DEBUG] ADMM Input - Fuel Efficiency: 5.0
[DEBUG] ADMM Allocation: {'Route A': 1, 'Route B': 7, 'Route C': 2}
[DEBUG] Idle Buses: 0
[OPTIMIZER] Updated fleet status.
[DEBUG] Fleet assignment: {'routes': {'Route A': 1, 'Route B': 7, 'Route C': 2}, 'idle_buses': 0}
[DEBUG] Passenger counts reset to zero
[DEBUG] Received data: {'counts': [1, 4, 5, 42, 9, 9]}
[DEBUG] Reset signal sent to client
[DEBUG] Updated passenger counts: [0, 1, 4, 5, 42, 9, 9, 0]
127.0.0.1 - - [15/Apr/2025 11:28:40] "POST /update_counts HTTP/1.1" 200 -
[DEBUG] Fleet Status Response: {'routes': {'Route A': 1, 'Route B': 7, 'Route C': 2}, 'idle_buses': 0, 'passenger_counts': [1, 4, 5, 42, 9, 9]}
127.0.0.1 - - [15/Apr/2025 11:28:40] "GET /fleet_status HTTP/1.1" 200 -
[DEBUG] Received data: {'counts': [0, 0, 0, 0, 0, 0, 0]}
[DEBUG] Updated passenger counts: [0, 0, 0, 0, 0, 0, 0, 0]
127.0.0.1 - - [15/Apr/2025 11:28:45] "POST /update_counts HTTP/1.1" 200 -
[DEBUG] Fleet Status Response: {'routes': {'Route A': 1, 'Route B': 7, 'Route C': 2}, 'idle_buses': 0, 'passenger_counts': [0, 0, 0, 0, 0, 0, 0]}
127.0.0.1 - - [15/Apr/2025 11:28:46] "GET /fleet_status HTTP/1.1" 200 -
```

Figure: ADMM Optimization Output

ADMM based Optimization

- To handle the large-scale routing optimization, the Alternating Direction Method of Multipliers (ADMM) was employed. The method breaks the global problem into smaller convex subproblems, enabling efficient parallel computation. This helped in generating optimal bus routes that adapt to demand fluctuations in real time.

Hardware Setup for Passenger Detection

- The hardware system comprises an Arduino Mega and IR sensors placed at bus entry points to count passengers in real time. This data is forwarded for model input and monitoring

Results and Discussion

```
Stop 1 → 0 passengers
Stop 2 → 1 passengers
Stop 3 → 3 passengers
Stop 4 → 1 passengers
Stop 5 → 1 passengers
Stop 6 → 1 passengers
[DEBUG] Raw data from Arduino: DATA:0,1,3,1,1,1
Stop 1 → 0 passengers
Stop 2 → 1 passengers
Stop 3 → 3 passengers
Stop 4 → 1 passengers
Stop 5 → 1 passengers
Stop 6 → 1 passengers
[DEBUG] Raw data from Arduino: DATA:0,1,4,1,1,1
Stop 1 → 0 passengers
Stop 2 → 1 passengers
Stop 3 → 4 passengers
Stop 4 → 1 passengers
Stop 5 → 1 passengers
Stop 6 → 1 passengers
```

Figure: Passenger Count Display Output

Results and Discussion

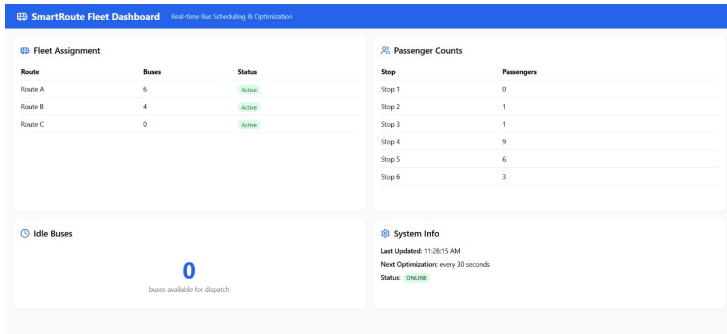


Figure: ReactJS Front-End Dashboard

Conclusion

SmartRoute ADMM provides an adaptive solution to urban bus scheduling and resource allocation. By combining AI-driven demand forecasting with ADMM-based optimization, the system enables real-time, demand-responsive scheduling, enhancing passenger experience, service reliability, and operational efficiency. Its scalable network supports dynamic routing, making it ideal for fluctuating urban transit demands

For future work, we aim to:

- Improve forecasting algorithms using advanced machine learning techniques.
- Explore integration with autonomous vehicles for fully automated operations
- Incorporate real-time data (weather, traffic, events) for further optimization.
- Expand to include multi-modal transport systems (bikes, metro, etc.).
- Test scalability to handle larger urban areas and more complex routes
- Optimize energy efficiency with sustainable technologies like electric buses.

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Questions?



Feel free to ask anything!

Thank You!

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We appreciate your time and attention!